

Data Flow Diagram

Date	15 March 2026
Team ID	XXXXXX
Project Name	OPTICROP – SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE
Maximum Marks	2 Marks

Data Flow Diagram

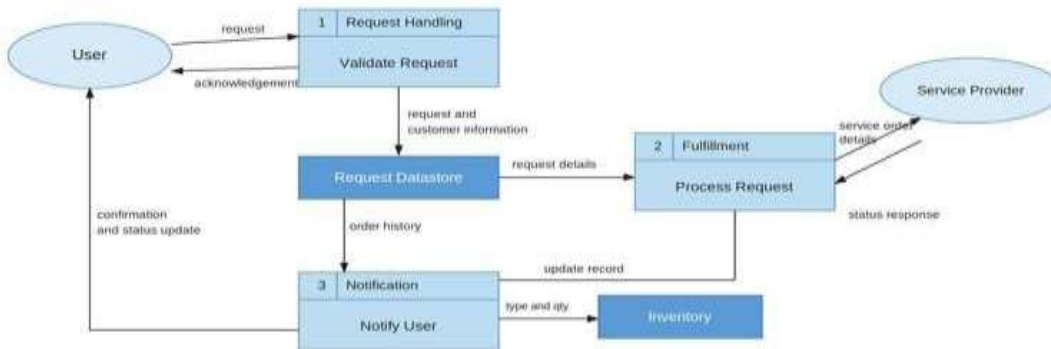
Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

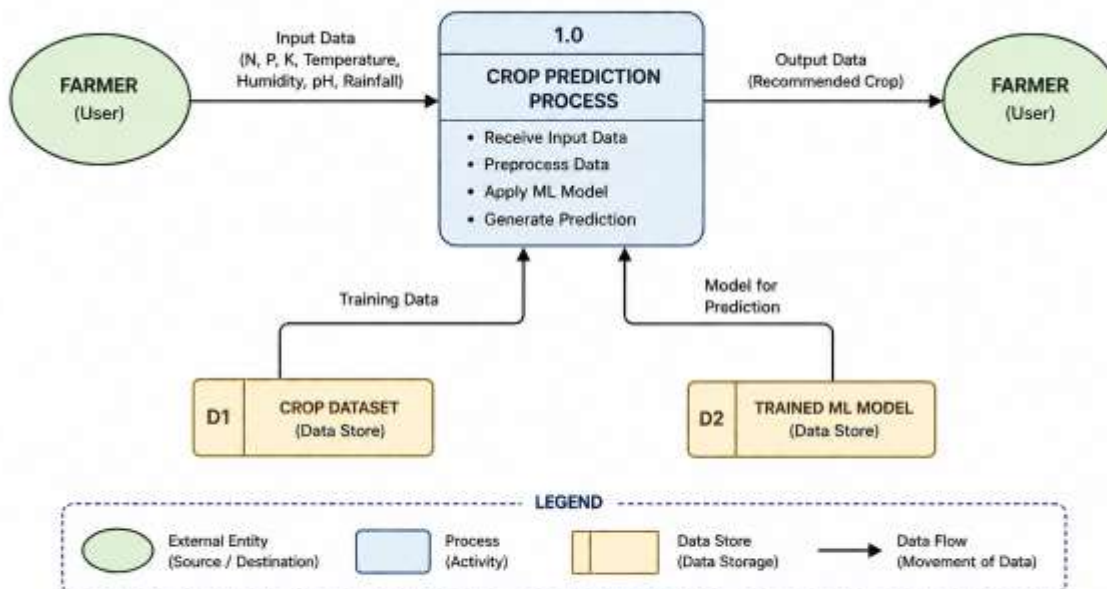
Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.



DATA FLOW DIAGRAM (DFD - LEVEL 0)



EXPLANATION:

- Farmer provides soil nutrient values (N, P, K) and weather parameters (Temperature, Humidity, pH, Rainfall).
- The system uses the Crop Dataset and Trained ML Model to generate the best suitable crop.
- The system returns the Recommended Crop to the farmer.